

Linjun SHOU

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Partner Group Applied Scientist Manager currently leading the AI efforts of Edge browser and office agent consisting of 30+ engineers/scientists. Linjun also has extensive experience in search technologies, including ranking, document understanding, query understanding (QU), and Q&A systems. He has published dozens of papers in top-tier research conferences and holds more than 20 U.S. patents. Additionally, Linjun is also a two-time recipient of the Distinguished Contribution Award offered by the CTO office at the Microsoft MLADS conference in both Spring and Fall of 2021.

Expertise: **Machine Learning, Copilot, Edge Intelligence, Document Understanding, Search, Question Answering, Recommendation, Natural Language Processing** etc.

Education & Employment

EdgeML and SocietasML Team, Microsoft Redmond

Partner Group Applied Science Manager

2025-Present

Edge Machine Learning, Microsoft China

Principal Group Applied Scientist Manager

2022-2025

Bing Search, Microsoft STCA

Principal Applied Scientist Manager.

2015-2022

Institute of Computing Technology, Chinese Academy of Sciences

Master's degree. Computer Sciences (Honored graduate).

2012-2015

Shanghai University

Bachelor's degree. Mathematics (**Top 1/297** in 4 consecutive years).

2018-2012

Work & Projects

Copilot in Edge Browser

Edge ML, Microsoft

2023/03-2024/05

- Motivation & Goal: Lead the team to build the most intelligent browser with Edge copilot.
- Accountability: Fully responsive for the Edge copilot algorithm, feature development including 1) Contextual chat with any documents in browser 2) Video copilot - chat with any videos 3) Edge actions for e2e task completion 4) Personal copilot - ask anything about what you have browsed before.
- Media report: [link1](#), [link2](#), [link3](#)
- Demo videos: [video1](#), [video2](#), [video3](#)
- Deliveries: (1) **Chat DAU reaches 3 million (10x increase)** (2) Contextual chat supports 1000+ pages long documents with 95%+ accuracy
- (3) **From Microsoft CEO Satya "my most favorite agenic experience is Tab auto grouping in Edge copilot"** (4) Positive user feedbacks (e.g. "God bless those who invented Edge copilot" from twitter user).

Copilot in Windows

Edge ML, Microsoft

2023/03-2024/05

- Motivation & Goal: To make sure the agile shipping of the first Windows copilot release, Edge team joint force with Windows team. Windows copilot is built on top of the Edge copilot code base.
- Responsibility: My team is responsible for the back-end and algorithm development to make sure the epical shipping of the copilot in the world's largest operation system.
- Deliveries: (1) Shipped the **first Windows copilot release** with positive user feedbacks (2) Shipped Windows skills which are powered by the Edge action framework I build. (3) Shipped the contextual chat in Windows copilot (4) Shipped the screen understanding feature - See what you see on your screen through screenshot.

Edge Client Intelligence - Client DNN models

Edge ML, Microsoft

2022/06-2023/03

- Developing a series of extreme model compression technologies including distillation and quantization etc to ship multiple client DNN models (DeepEE, DeepPU, DeepAutoFill, DeepUU) to power various AI features in Edge with enhanced privacy and cogs saving.
- DeepEE - **Shipped the very first client universal transformer models in Edge client** w/ P/R over 95 to 69 languages, 154+ regions world-wide for PH, PC to power Shopping globalization. 4 million peak DAU in shipping season
- DeepPU - shipped to 6+ scenarios to power shoreline personalization, 1CNS personalized notifications, contextual prompt suggestions, shopping PDP triggering, tab group autonaming and auto-grouping, etc
- DeepAutoFill - auto filling scenarios in Edge
- DeepUU - Shipped V1 to build the client user profile in global markets, powering the Edge personalization w/ BSoM proxy gains +0.79% (abs) and +2% (rel) .

Document Understanding for 400b Web Index

Document Understanding Team, WebXT, Microsoft

2019-2022

- Lead STCA DU team (16 engineers/scientists) to develop the best document understanding DNN techniques and platform for search & recommendation scenarios.
- Develop **Universal Multi-modal Information Extraction technology** (Semantic Document) to power 15+ prod scenarios, including Caption, QnA, Ads, Recommendation, etc. Further expand the technique to support fine-grained annotation extraction, such as product name, price, manufacture, image, etc.
- Develop **Universal Topic Understanding models with 100k topics** to power 15+ scenarios including Sage News, Windows Feeds (50+ global regions), Underside, Enterprise News, Bing QnA, PAA, etc.
- Develop SOTA document full body representation model (FBV) - shipped to Bing search for **400 billion vector space ANN search**.
- Develop Long Document MRC: one of the longest DU model in 100+ languages to power 8+ scenarios, including Azure, QnA, PAA, MSN News, Knowledge Card, etc.
- Develop authority techniques (topical authority and host embedding) to power Bing QnA, PAA, core relevance, spam detection, personalized search, etc.
- Excellent team collaboration capabilities - receive **16 positive partner feedbacks (VP, Partner, Principal level) in FY21** across multiple orgs.

Recommendation Algorithms for Feeds

ASK & DU Team, Bing, Microsoft

2020-2020

- QnAFeeds - as tech lead to drive the recommendation streaming indexgen, ranking system (3 layers of ranking and two-tower embedding model) from scratch.
- News Recommendation - develop and ship topic/KPE based personalized recall, topic based diversity to support **the first Windows Feeds release in 50+ regions on Windows 10/11 system**.
- Web Recommendation - as tech lead to drive the FBV techniques for semantic recall path, which achieves **21 points recall@100 gains**.
- Movie Recommendation - develop knowledge graph enhanced collaborative filtering algorithm to achieve 10 points gains.

Bing QnA Universalization in Global Regions

ASK Team, Bing, Microsoft

2019-2020

- As tech lead to **scale QnA from English only to 100+ languages and 200+ regions** for the very first time.
- Propose a systematic language scaling methodology to resolve the training data limitation challenge on low-resource languages [10, 16, 11, 13, 7]. Two works about language scaling won **Distinguished Contribution Award of Microsoft MLADS conference twice** (top 2/300, record-breaking).
- Design and drive the development of a brand-new universal streaming QnA pipeline from scratch and improve the QnA refresh circle from **1 months to minutes/seconds** and **save 80%** of the GPU capacity.

Deep Learning Technologies for Bing QnA

Quantus Team, Bing, Microsoft

2017-2019

- Ship **10+ releases DNN techniques** to improve the core QnA relevance, which beat and replace legacy GBDT based ranker with 400+ handcrafted features for the first time and win WPSBS with Google.
- Design and build DNN powered full stack list QnA ranking pipeline and ship **10+ releases**, which contribute to **20+ precision gains** to beat Google.
- Propose novel model compression techniques for heavy pre-trained model online serving [**Shou2020RSKD**, 15] which powered **the first BERT model released in Bing** (within 1 month of BERT's open source) with **3.0% coverage gains** and **5.0% precision gains** for QnA.
- Summarize the model training experience internally and open-source a NLP DNN toolkit called *Neuron-Blocks* on Github, which gets **1k+ stars in 2 months** [17]. DNN model development time can be reduced from **weekly/daily to minutely**.

Bing News and Tweet Core Ranking

News Team, Bing, Microsoft

2015-2016

- Ship dozens of ranker releases in Bing news ranking and News homepage ranking.
- Ship ML based Tweets ranker that greatly improved the twitter answer defect rate from **18% to 2%**.
- Design breaking news detection system to achieve minute-level latency (from daily) and file a US patent.
- Develop and ship the RichNav news feature on Bing from scratch, which leads to **2.73 strong surplus(SS)**, **19.17 weak surplus(SS)**.

Selected Publications

Published **50+ papers** in top tier conferences with **9500+ citations** on Google scholar. Full publications can be found on [my scholar homepage](#).

- [1] Nuo Chen#, **Linjun Shou**, Tengtao Song, Ming Gong, Jian Pei, Jianhui Chang, Daxin Jiang, and Jia Li. "Structural Contrastive Pretraining for Cross-Lingual Comprehension". In: *ACL*. 2023 (# intern I mentored).
- [2] Nuo Chen#, **Linjun Shou**, Ming Gong, Jian Pei, Bowen Cao, Jianhui Chang, Daxin Jiang, and Jia Li. "Alleviating Over-smoothing for Unsupervised Sentence Representation". In: *ACL*. 2023 (# intern I mentored).
- [3] Shengyao Zhuang#, **Linjun Shou**, Jian Pei, Ming Gong, Houxing Ren, G. Zuccon, and Daxin Jiang. "Typo-aware Bottlenecked Pre-Training for Robust Dense Retrieval". In: *SIGIR*. 2023 (# intern I mentored).
- [4] Nuo# Chen, **Shou, Linjun**, Ming Gong, Jian Pei, and Daxin Jiang. "Bridging the Gap between Language Models and Cross-Lingual Sequence Labeling". In: *NAACL*. 2022 (# intern I mentored).
- [5] **Linjun Shou**, Ming Gong, Jian Pei, Xiubo Geng, Xingjie Zhou, and Daxin Jiang. "Language Scaling: Applications, Challenges and Approaches". In: *KDD Tutorial*. 2021.
- [6] **Linjun Shou**, Ming Gong, J. Pei, Xiubo Geng, Xingjie Zhou, and Daxin Jiang. "Scaling out NLP Applications to 100+ Languages". In: *WWW Tutorial*. 2021.
- [7] S. Liang #, M. Gong, J. Pei, **Linjun Shou**, and D. Jiang. "Reinforced Iterative Knowledge Distillation for Cross-Lingual Named Entity Recognition". In: *KDD*. 2021 (# intern I mentored).
- [8] Yingmei Guo, **Linjun Shou**, Jian Pei, Ming Gong, Mingxing Xu, Zhiyong Wu, and Daxin Jiang. "Learning from Multiple Noisy Augmented Data Sets for Better Cross-Lingual Spoken Language Understanding". In: *EMNLP*. 2021.
- [9] Junjie Huang, Duyu Tang, **Linjun Shou**, Ming Gong, Ke Xu, Daxin Jiang, and Ming Zhou. "CoSQA: 20,000+ Web Queries for Code Search and Question Answering". In: *ACL*. 2021.
- [10] **Linjun Shou**, S. Bo, Fei-Xiang Cheng, Ming Gong, J. Pei, and Daxin Jiang. "Mining Implicit Relevance Feedback from User Behavior for Web Question Answering". In: *KDD*. 2020.
- [11] Shining Liang, **Linjun Shou**, Jian Pei, Ming Gong, Wanli Zuo, and Daxin Jiang. "CalibreNet: Calibration Networks for Multilingual Sequence Labeling". In: *WSDM*. 2021 (# intern I mentored).
- [12] Junwei Liao, Yu Shi, Ming Gong, **Linjun Shou**, Sefik Eskimez, L. Lu, H. Qu, and Michael Zeng. "Generating Human Readable Transcript for Automatic Speech Recognition with Pre-trained Language Model". In: *ICASSP*. 2021.
- [13] Junhao Liu*, **Linjun Shou***, Jian Pei, Ming Gong, Min Yang, and Daxin Jiang (* Equal contribution). "Cross-lingual Machine Reading Comprehension with Language Branch Knowledge Distillation". In: *COLING*. 2020.
- [14] F. Yuan#, **Linjun Shou**, X. Bai, M. Gong, Y. Liang, N. Duan, Y. Fu, and Daxin Jiang. "Enhancing Answer Boundary Detection for Multilingual Machine Reading Comprehension". In: *ACL*. 2020 (# intern I mentored).

- [15] Ze Yang*, **Linjun Shou***, Ming Gong, Wutao Lin, and Daxin Jiang (* Equal contribution). "Model Compression with Two-stage Multi-teacher Knowledge Distillation for Web Question Answering System". In: *WSDM*. 2020.
- [16] H. Huang, Yaobo Liang, N. Duan, Ming Gong, **Linjun Shou**, D. Jiang, and M. Zhou. "Unicoder: A Universal Language Encoder by Pre-training with Multiple Cross-lingual Tasks". In: *EMNLP/IJCNLP*. 2019.
- [17] Ming Gong, **Linjun Shou**, Wutao Lin, Zhijie Sang, Quanjia Yan, Ze Yang, and D. Jiang. "NeuronBlocks - Building Your NLP DNN Models Like Playing Lego". In: *EMNLP/IJCNLP*. 2019.
- [18] Xuguang Wang, **Linjun Shou**, Ming Gong, Nan Duan, and Daxin Jiang. "No Answer is Better Than Wrong Answer: A Reflection Model for Document Level Machine Reading Comprehension". In: *EMNLP*. 2020.

Patents

- o Personalized Multilingual Content Recommendation based on Robust Feature Network 2023.(MS# 412659-CN-NP).
- o Split-Screen Viewing Based On Intelligent Triggering. 2023. MS# 413279-CN-NP. Sequential Recommendation based on Cross-Domain Behavior Data. 2023. MS# 412874-CN-NP.
- o Universal Deep Entity Extraction (DeepEE) for Browser. 2023.
- o Providing An Activeness Indication Of A Tab. 2023. MS# 412658-CN-NP.
- o ESTABLISHING A LANGUAGE MODEL ADAPTED TO A CROSS-LINGUAL SEQUENCE LABELING TASK. 2022. MS# 411268-CN-NP.
- o Temporal Co-Contrastive Learning-Based Node Representation Generation. 2022. MS# 411267-CN-NP.
- o Sequential Recommendation Based On Dual Contrastive Interest Learning. 2022. MS# 411269-CN-NP.
- o Sentence Representation Generation for Cross-lingual Retrieval. 2022. MS# 411088-CN-NP.
- o Content Recommendation Based On Graph Enhanced Collaborative Filtering. 2022. MS# 410901-CN-NP.
- o Controversial Poll Auto Generation by Generation Model and Crowdsourcing. 2021.
- o Knowledge enhanced Content-Aware Collaborative Filtering for Recommendation. 2021.
- o From Good to Best: Two-Stage Training for Cross-lingual Machine Reading Comprehension. 2021.
- o Graph based Fact Clustering for Diversity and Freshness. 2021.
- o Representation Learning of Cross-Language Texts. 2021. MS# 409565-CN-NP
- o Hierarchical Representation Learning of User Interest. 2021. MS# 410462-CN-NP
- o Question Generation from Queries. 2021. MS# 410241-CN-NP
- o Multi-Model Joint Denoising Training. 2021. MS# 409566-CN-NP.
- o Representation Learning of Semi-structured Data. 2020. MS#: 408908-CN-NP.
- o Model Selection Learning for Knowledge Distillation. 2020. MS# 408426-CN-NP.
- o Providing QA Training Data and Training a QA Model based on Implicit Relevance Feedbacks. 2020. MS# 407927-CN-NP.
- o Contrastive Learning for Question Answering. 2020. MS# 407988-CN-NP.
- o Model Generation based Model Compression. 2019. MS# 406805-CN-NP.
- o Cross-Lingual Task Training. 2019. MS# 406497-CN-NP.
- o Assertion-based Question Answering with Question-Aware Open Information Extraction. 2019.
- o Event Factory: detect event's scale, subintent, category, timeline, social sentiment with artificial intelligence for search and recommendation. 2016. US WO2018098751A1.

Awards

- o Microsoft 2024 Hackathon **Greater China Region Outstanding Project Award**. Bootstrap Copilot Pages with Edge Browsing Task Center [link](#). This project won 4 awards in total which is record-breaking.
- o Microsoft 2024 Hackathon **Beijing Outstanding Project Award**. Browse for Me [link](#).
- o Microsoft 2024 Hackathon **Beijing Outstanding Project Award**. Microsoft Mindmap - Transform Anything into Mind Maps [link](#).
- o Microsoft 2024 Hackathon **Suzhou Outstanding Project Award**. Tango - On-Device AI Service Stack for Windows M365 Apps [link](#).

- Microsoft 2021 **MLADS Fall Distinguished Contribution Award (2/250+)**. The first time in history to receive the Award twice in 2 consecutive conferences.
- Microsoft 2021 **MLADS Spring Distinguished Contribution Award (2/300+)**.
- Excellent Serviced AS Teaching Assistant of AI School China. 2019, 2020, 2021 (Won Excellent Project Award - 1st Prize twice).
- Microsoft AI Core Q3 FY18 Greatness Award. 2018.
- Excellent Student of CAS (Top 5%). 2013.
- Honored Graduate of Shanghai (Top 1%). 2012.
- National Scholarship (Top 1%). 2011.
- First Prize Scholarship in SHU (Top 1%) (Four consecutive times). 2008-2012.